BABARINDE JOHNSON OMOTAYO

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Ibadan, Oyo State, Nigeria | Open to Remote (Worldwide)

PROFESSIONAL SUMMARY

AI Engineer specializing in Large Language Models (LLMs), Natural Language Processing, and low-resource language AI systems. Experienced in building and fine-tuning transformer-based models, including GPT-style architectures, and deploying real-world AI products. Creator of Odùduwà AI, a live African language intelligence platform for Yorùbá NLP. Proven ability to develop end-to-end AI systems from model training to production deployment. Strong focus on model optimization (LoRA, GGUF) and scalable AI solutions. Passionate about advancing generative AI, speech systems, and applied machine learning for real-world impact.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (React, Node.js)
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain
- **Data & Visualization:** NumPy, Pandas, Matplotlib, Power BI
- **AI/ML & LLM Engineering:** Transformers, GPT architectures, LLM fine-tuning, LoRA, GGUF optimization, NLP, CNNs, RNNs, Wav2Vec2.0
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle, Power BI, LangChain
- **Soft Skills:** Problem-solving, remote collaboration, project ownership, leadership, adaptability

PROJECTS

GPT-2 Language Model (From Scratch) — 2025

- Developed a mini GPT-2 implementation in PyTorch from first principles, including tokenizer, multi-head attention, and training loop for text generation tasks.

Odùduwà AI — African Language Intelligence Platform | 2025 | Live: oduduwaai.netlify.app

- Built and deployed a full-stack AI platform for Yorùbá language processing
- Developed features including: AI conversational assistant, Yorùbá–English translation with cultural context, tone/diacritic restoration system

Readly (ReadlyAI) — Document Q&A Web App | 2025 | Live: readly-doc.vercel.app

- Built and deployed a full-stack RAG application enabling users to upload documents (PDF/DOCX/TXT) and ask questions about their content
- Implemented retrieval pipeline using FAISS vector search with Cohere embeddings, powered by Groq LLM for fast inference
- Deployed backend on Render (readly-rag.onrender.com) and frontend on Vercel with a custom domain

Yorùbá LLM Models (Published on Hugging Face) — 2025

- Fine-tuned multiple transformer models (LLaMA, Qwen) for Yorùbá NLP tasks
- Implemented LoRA-based parameter-efficient training
- Converted models to GGUF format for efficient local inference
- Developed models for: Conversational AI, Translation (Yorùbá ↔ English), Language understanding

OMR Detection System (YOLO + Flask API) — 2025

- Developed an automated Optical Mark Recognition system using YOLO
- Built REST API using Flask for real-time processing
- Applied image preprocessing and detection pipelines for high accuracy

DeepSeek Transformer — 2025

- Implemented and trained a DeepSeek-style transformer model using PyTorch for contextual sequence modeling; optimized with gradient clipping and mixed precision training.

Wav2Vec2.0 Voice Assistant (“Pedia”) — 2024

- Designed a custom voice assistant using Wav2Vec2.0 for speech recognition integrated with Python's text-to-speech and NLP pipeline for conversational response generation.

Agricultural Image Classifier — 2023

- Built CNN models for crop disease classification using TensorFlow; achieved 91% validation accuracy on an agricultural dataset.

PUBLICATIONS

Babarinde, J. O., Nath, A., Nath, A. A Web Based Automated OMR Evaluation System Using YOLO and Image Processing Techniques. International Journal of Engineering Research and Technology (IJERT), Vol. 14, Issue 12, December 2025. ISSN: 2278-0181.

EXPERIENCE

Independent AI & Machine Learning Developer | Remote | 2023 – Present

- Built and trained Transformer-based architectures (GPT-2, DeepSeek) from scratch
- Developed and deployed CNNs and speech-to-text models for research projects
- Applied machine learning to agri-tech and food data for predictive analysis

Python Programming Tutor | Dept. of Food Science, LAUTECH | Ogbomoso, Nigeria | 2021 – 2024

- Taught and mentored students in Python programming and algorithmic problem-solving
- Designed coding tutorials and guided over 100 learners through ML basics

EDUCATION

B.Tech. in Food Science | Ladoke Akintola University of Technology (LAUTECH) | Ogbomoso, Nigeria | 2024

CGPA: 3.94 / 5.00 — Second Class Upper Division

Data Research Intern | International Institute of Tropical Agriculture (IITA) | 2025

Data Research and Methods Unit

- Supported data-driven research through data collection, cleaning, analysis, and visualization for agricultural studies
- Applied statistical and computational methods to analyze structured research datasets and interpret results
- Gained hands-on experience with research workflows, reproducibility, and collaborative scientific analysis
- Worked alongside multidisciplinary researchers, strengthening skills in data reasoning and applied research methodology

CERTIFICATIONS & ACHIEVEMENTS

- Overview of Artificial Intelligence — Huawei (2025)
- Basics of Machine Learning Algorithms — UniAthena (2025)
- Python Coder — Kaggle (2025)
- Jobberman Soft Skills Training (2023)
- Kaggle Top 99 Accuracy (2025)
- Certificate of Publication — IJERT (2025)